

3.2.2 Design 2 (Figure B.2)

Radiologists diagnose each patient case in each of the four information environments in the second design. For the moment, set aside concerns arising from the feature that the same radiologist encounters the same case multiple times. This design will allow us to estimate an empirical analog to equation (5). It also has the added benefit of controlling for both case-radiologist heterogeneity because, unlike in the previous design, we can conduct within-case-radiologist comparisons across treatments.

Because radiologists repeatedly encounter cases, we need to address the potential for order effects due to memory. For example, radiologists might anchor on their previous assessment using AI predictions or contextual information and might remember this information the next time the same case is encountered. We, therefore, limit radiologists’ ability to remember either their diagnosis or previously provided information by using a “washout” interval between two encounters of the same case.¹² Specifically, radiologists complete the experiment in four sessions that are separated by at least two weeks. Each session is similar to the first design: radiologists diagnose fifteen cases in each of the four information environments with no case repeated within a session. Across sessions, the information environment under which a given case is diagnosed is permuted. Thus, by the end of the fourth session, each of the sixty cases is diagnosed exactly once in each information environment. Our results are consistent with the washout being effective – radiologists’ predictions do not move towards the AI prediction if it was provided in a prior session but do if it is provided in the current session (see figure C.28).

3.2.3 Design 3 (Figure B.3)

In the third design, we address residual concerns about the order effects of radiologists diagnosing cases with AI before those without AI—whether due to anchoring, memory, or experimenter demand—by having participants diagnose fifty cases, first without and then with AI assistance. Within each block, clinical history is randomly provided in either the first or second half of images.

This design also allows us to conduct within case-radiologist comparisons. The potential disadvantage of this design is that we cannot distinguish order effects from the effect of providing AI. This issue is unavoidable given the guiding principle that participants receive weakly more information about a case during a repeat encounter. However, we can test for

¹²This principle has been used in computer science (Seah et al., 2021; Conant et al., 2019; Pacilè et al., 2020).

and do rule out order effects on accuracy based on the first two designs.

3.2.4 Participant Recruitment

Participants for the first and third designs, which constitute the majority, were recruited through teleradiology companies. The teleradiology companies allow us to recruit several experts in a relatively liquid spot market, a practice that is now common for decision experiments with non-expert subjects (Hunt and Scheetz, 2019). Most healthcare providers in the US rely on these companies’ services, although many large hospitals have on-call radiologists (Rosenkrantz et al., 2019). We work with teleradiology companies that serve US hospitals and offer the services of both US-based and non-US-based radiologists. Our contracts with teleradiology companies specify a piece-rate, and the companies, in turn, compensate the participants with a piece-rate.¹³ In addition, we provided monetary incentives for accuracy to a subset of radiologists, as described in the next section.

The second design required us to work with a partner who could guarantee subjects’ participation over several months. We collaborated with VinMac healthcare system in Vietnam to recruit their staff radiologists to ensure continued participation. VinMac is in the process of developing its own in-house AI capabilities and was willing to assist with our experiment in exchange for recognition in a publication of the resulting dataset. The VinMac radiologists did not receive any payments to participate in the experiment but we find that their performance is very close to the performance of the tele-radiologists.

In total, 180 radiologists participated in our experiment. Close to 17% of our participants are US-based, 38% have a degree from a US institution, 80% are affiliated with a large clinic, and 61% with an academic institution. As demonstrated in appendix C.3, the quality of the assessments made by the radiologists in our study is comparable to that of the staff radiologists from Stanford University Hospital, who originally diagnosed the patient case.

3.2.5 Incentives

We cross-randomize incentives for accuracy in the first and third designs but not the second because of the specific ways in which our partner’s radiologists are employed.¹⁴ Payments were determined following the binarized scoring rule in Hossain and Okui (2013), where truth

¹³We worked with three companies with piece-rates ranging from \$7.50 to \$13.00.

¹⁴There does not appear to be a consensus in the experimental literature on whether incentives are superior to non-incentivized responses when eliciting beliefs (see Danz, Vesterlund, and Wilson, 2020; Charness, Gneezy, and Rasocho, 2021). A comparison between the incentivized and non-incentivized participants allows us to answer this question in the context of this experiment.

is determined as described in section 3.3.1 below. This incentive scheme uses a loss function of the mean squared prediction error, averaging over patient cases and pathologies, and the respondents earn a fixed bonus of \$120 if a random draw is less than the loss function. This bonus is more than 20% of the base payment to teleradiology firms. We explain to the participants that expected payments are maximized if they provide their best estimates using a non-mathematical description of the payment rule. We specify the distribution so that 30% of pilot participants would earn the bonus, cross-randomized with the other two treatment arms.

3.3 Implementation and Data Collection

3.3.1 Patient Cases and Ground Truth

The experiment uses 324 historical patient cases with potential thoracic pathologies from Stanford University’s healthcare system. For each case, we have access to the chest X-ray and the clinical history in the form of the primary provider’s written notes, the patient vitals, and demographics.¹⁵ The use of retrospective cases allows us to avoid ethical and other issues that would arise when experimenting in high-stakes settings.

Our analysis requires constructing ω_i for each patient case. There are important challenges in using an observational dataset of patient health records to construct this field. One approach would be to use the results from further medical tests. Unfortunately, definitive gold-standard tests do not exist for most thoracic pathologies.¹⁶ Even when follow-up tests are conducted, they are selected on the likely presence of a pathology, an issue referred to as the selective labels problem (e.g. Mullainathan and Obermeyer, 2019). Medical outcomes from patient health records also do not suffice because actions taken by the treating physician in response to the radiology report contaminate these measures. Recent literature has suggested instrumental variables approaches for solving this selective labels problem, but this work targets population quantities and not a “ground truth” on each case (e.g. Chan et al., 2022; Mullainathan and Obermeyer, 2019).

We construct ω_i by aggregating the assessment of a group of expert radiologists, an approach common in computer science (Sheng et al., 2008; Mccluskey et al., 2021). Specifically, we

¹⁵All cases are first encounters with no prior X-ray as a comparison. We started with 500 cases that fit these primary criteria. We omitted pediatric cases from this set. Finally, a radiologist reviewed the cases to remove instances with poor image quality. The clinical history was manually reviewed to remove patient-identifiable information and cleared for public release.

¹⁶Many pathologies do not have commonly used non-imaging-based diagnostic tools. For instance, the presence of cardiomegaly – an enlarged heart – can only be determined using imaging tools, thoracic surgery, or an autopsy.

ask five board certified radiologists from Mount Sinai to read each of the 324 cases using the interface described above with the available X-ray and clinical history. For each case i and expert radiologist r , we, therefore, obtain $\pi_r(\omega_i = 1 | s_{i,r}^E)$ for each pathology, which we aggregate to generate ω_i . Specifically, we classify $\omega_i = 1$ if $\sum_r \pi_r(\omega_i = 1 | s_{i,r}^E) / 5 > 0.5$. This approach immediately addresses the selective labels problem because the availability of assessments is not selected on the likelihood of a pathology being present. Results in [Wallsten and Diederich \(2001\)](#) suggest that, under weak conditions that allow for measurement error in the reports and correlations across reports, the aggregate opinion of several experts is highly diagnostic as long as the experts are median unbiased. To assess robustness, we aggregate $\pi_r(\omega_i = 1 | s_{i,r}^E)$ using both a log-odds average and a leave-one-out mean of the assessments of all the radiologists in our experiment in the clinical history treatment condition.

3.3.2 Experimental Interface and Data Collected

We developed the interface to present the patient cases and to collect radiologists’ predictions and decisions in collaboration with board certified radiologists at Stanford University Hospital and Mt. Sinai Hospital. In contrast to free-text reports, we designed it to generate structured and quantitative data on chest X-ray responses that resemble a typical radiological report. A guiding principle in the design was to mimic clinical practice and to present and obtain all clinically relevant information. We briefly describe this interface and provide images and further details in appendix [B](#).

On the landing page of each case, a high-resolution image of a patient’s X-ray is presented to the radiologist, with the functionality to zoom and adjust brightness and contrast. When the experiment calls to show the clinical history, the interface presents clinical notes, vitals, and laboratory results available at the time the X-ray was originally ordered. If the experiment provides AI assistance, participants are shown AI predictions for fourteen pathologies and a composite prediction for whether or not there are any relevant findings.

The data entry interface collects a radiologist’s assessments of the probability that various thoracic pathologies are present for a given case. The probability that a pathology is present given the available information, i.e. $p_r(\omega = 1 | s)$, is elicited using a continuous slider. We visually subdivide possible responses into five intervals with standard language labels used in written radiological reports to aid the participants.¹⁷ Our radiology collaborators grouped pathologies into eight exclusive parent categories based on their type. Each group has children

¹⁷The specific labels are “Not present”, “Very Likely”, “Unlikely”, “Possible”, “Likely”, and “Highly Likely”. Several radiological publications have suggested such standardized language for radiological reports. See for instance [Panicek and Hricak \(2016\)](#).